

Charging Station Placement for Limited Energy Robots^{*}

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Abstract

We address the *minimum charging station (MIN-STATION) placement* problem for unlabelled robots. The input consists of a graph $G = (V, E)$, with given starting and target positions $S \subset V$ and $T \subset V$, respectively, for a set of m robots. A robot moves one step by transitioning from one vertex to an adjacent vertex in G and has a movement capacity of r steps, meaning it can move up to r steps without recharging from either its starting position or a charging station. Each target position must be occupied by exactly one robot to complete the motion, hence $|S| = |T|$. The objective is to determine the minimum number of charging stations needed to ensure all target positions are reached. We prove that this problem is NP-hard, even for bounded degree graphs with a maximum degree of 6. However, on the positive side, we present linear and quadratic time algorithms for the MIN-STATION problem on paths and cycles.

Keywords: Minimum station placement, Robot motion planning, Resource allocation problem, Computational geometry.

1. Introduction

Robot motion planning [1] has emerged as a prominent area of research, driven by its fundamental significance in the advancement of robotics and automation technologies. Multiple mobile robots move together in order to complete one or multiple tasks in warehouses management [2], airport

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operation [3], agricultural fields [4], military areas [5], and medical facilities [6]. The theoretical aspects of robot motion planning consists of several factors like *make span minimization* [7], *obstacle avoidance* [8], and *charger placement* [9]. In the classical model of robot motion planning, the robots commence their operation from a set of predetermined initial positions and navigate to some specified target locations to execute the designated tasks assigned to them. However, in practical scenario, robots commonly operate with limited (battery) energy capacity, necessitating periodic recharging to maintain functionality. Thus, the strategic placement of charging stations within the operational environment is essential, and it is crucial to minimize the number of such stations to achieve cost efficiency. In recent years, numerous approaches have been studied by researchers to mitigate energy constraints using docking based recharging [10], solar charging [11], and continuous tethering [12].

In this article, we consider the variant of the *minimum charging station placement* problem with unlabeled robots. The starting and target positions for a set of m robots on the vertices of a graph G are given as a part of the input, along with a positive integer r . A robot takes one step by moving from one vertex to an adjacent vertex in G . Each robot has a *movement capacity* of r steps. This means it can move r steps without recharging when starting from its starting position or a charging station. To complete the task assigned to the robots, each target position needs to be occupied by exactly one robot. We say that the *motion of the system* is completed when each target position is occupied by a robot. Throughout the paper, we assume that G is a simple connected undirected graph. This ensures the reachability of each target from each source. The goal of the problem is to compute the minimum number of charging stations required to complete the motion of the system, formally stated as follows.

Problem 1. MIN-STATION

Input: An undirected simple connected graph $G = (V, E)$, a set $S \subset V$ as initial positions of m robots, a set $T \subset V$ as target positions of the robots, such that $|S| = |T| = m$, and a positive integer r , as the movement capacity of the robots.

Output: A minimum cardinality subset C of V such that the robots can complete the motion of the system by reaching T from S , as defined earlier.

Since the robots are unlabeled, they are considered to be identical. So, there is no fixed matching between the starting positions and the target positions of the robots. At the completion of the motion of the system, exactly one robot must occupy each of the target positions. During the motion, multiple robots may arrive at one vertex simultaneously, allowing them to recharge concurrently at a single charging station. A vertex $v \in G$ is called a *terminal* if $v \in S \cup T$.

The literature addressing the recharge station placements for mobile robots has a pool of results involving different variants of the problems. Strimel and Veloso [13] considered the problem of placing minimum number of charging stations for a single robot movement with a limited battery. Kundu and Saha [9] studied the problem of finding the minimum number of charging stations for a group of mobile robots with battery capacity d where the *motion graph* of the robots are given as a part of the input. The motion graph is a directed graph which is the union of the directed paths traversed by the robot during the motion. Thus the problem converts to compute the minimum number of vertices in a directed graph such that each vertex is within a distance d from the computed set of vertices. They have also addressed the problem of minimizing d when the positions of a set of stations are given on the graph. They proved the NP-hardness of both problems and presented approximation algorithms. This model relates with the *center finding* problem which is widely studied in the *facility location* [14] problems. Beside theoretical results, heuristic algorithms [15] are also devised to address the problem of locating optimal set of charging stations on the motion graph of the robots. However, the models assume the paths of the robots to be the part of input when computing the optimal allocation of the charging stations.

We recall, in MIN-STATION, neither the target position for a specific robot nor its path are given as a part of the input. A robot may need to travel beyond its shortest paths from the starting and the target position. To the best of our knowledge, no result is present in the literature addressing MIN-STATION problem. We prove that MIN-STATION is NP-hard even for bounded degree graphs, where the maximum degree of the graph is bounded by 6. Then we present a linear time algorithm for MIN-STATION on paths followed by a quadratic time algorithm for cycles.

The article is organized in the following manner. We describe the necessary definitions in Section 2. Section 3 contains the NP-hardness result for MIN-STATION. Then we present two algorithms for MIN-STATION on paths and cycles in Section 4. Finally the article is concluded in Section 5 pointing

some immediate open problems arising from the work.

2. Preliminaries

We define some necessary terminologies in this section. We use most of the standard graph theoretic terminologies [16]. A graph $G = (V, E)$ is called a *path* if the vertices of G can be listed in order $V = \{v_1, v_2, \dots, v_n\}$ and the edges are (v_i, v_{i+1}) , for all $i = \{1, 2, \dots, n-1\}$. The *distance* between two vertices v_i and v_j of a path P consisting of the vertices $\{v_1, v_2, \dots, v_n\}$ is the absolute value of $(i - j)$.

A *cycle* is a path with $v_1 = v_n$. A cycle with n vertices is called an n -cycle. The distance between two vertices on a cycle is the number of edges present on the *shortest path* between the two vertices.

An *interval* $[x, y]$ of a path or a cycle is a set of consecutive vertices going from x to y in the vertex list of the path or cycle. We say that the interval is *spanned* by x and y . The *length of the interval* is the number of edges present in the subgraph *induced* by the vertices of the interval, denoted by $|x - y|$.

3. NP-hardness

We prove the NP-hardness of MIN-STATION by reducing the well-known NP-complete problem 3,3-SAT [17] into it. In this variant of the classical 3-SAT [18] formula each variable appears in at most three clauses either as a positive or as a negative literal.

Let us consider a 3,3-SAT formula ϕ consisting of n variables and m clauses. We note that in the problem definition, we use n, m as the number of vertices of the graph and the number of robots, respectively. However, in this section, these numbers do not conflict and are clear from the context.

We construct a graph G_ϕ with $(4n + 2m)$ vertices and $(m + n)$ robots such that MIN-STATION problem on G_ϕ has a solution of size n if and only if ϕ is satisfiable. We describe the construction of G_ϕ (depicted partially in Figure 1) as follows.

1. We introduce a 4-cycle for every variable x_i in ϕ , namely $v_{x_i}^1, v_{x_i}^2, v_{x_i}^3, v_{x_i}^4$. These cycles are referred to as *variable gadgets*.
2. We introduce two vertices $v_{c_j^1}$ and $v_{c_j^2}$ for each clause C_j in ϕ . These vertices are referred to as *clause gadgets*.

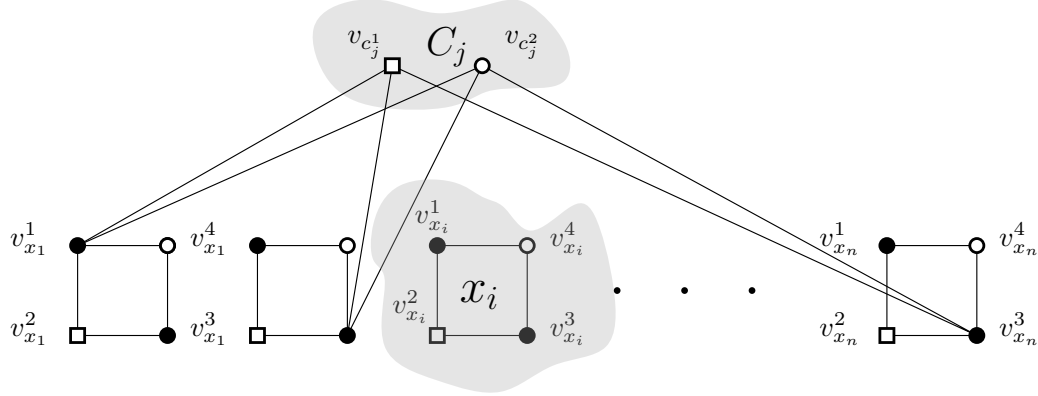


Figure 1: Formation of G_ϕ from a given formula ϕ . The connection of $C_j = (x_1 \vee \overline{x_2} \vee \overline{x_n})$ is depicted. The hollow squares and circles indicate the starting and target positions of the robots, respectively.

3. We introduce edges from $v_{c_j}^1$ and $v_{c_j}^2$ to $v_{x_i}^1$ or $v_{x_i}^3$ if C_j contains x_i as a positive literal or a negative literal, respectively.

Then we form an instance of MIN-STATION on G_ϕ where the set S of starting position of $(n + m)$ robots is $S = (\cup_{j=1}^m v_{c_j}^1) \cup (\cup_{i=1}^n v_{x_i}^2)$. The set T of target positions is $T = (\cup_{j=1}^m v_{c_j}^2) \cup (\cup_{i=1}^n v_{x_i}^4)$. The movement capacity of each robot is 1. Then we have the following lemma.

Lemma 1. *The solution of the MIN-STATION problem, with the input described above, is n on G_ϕ , if and only ϕ is satisfiable.*

Proof. First, let us assume that ϕ is satisfiable. Then at least one literal in each clause is assigned to be true. We can place the n stations in the vertices corresponding to the literals in the variable gadget. Then, for every clause C_j where $j = \{1, 2, \dots, m\}$, the robots starting in $v_{c_j}^1$ reach $v_{c_j}^2$ via the station placed at the vertex corresponding to the literal which is assigned as true. Since every variable gadget contains one station either at $v_{x_i}^1$ or at $v_{x_i}^3$, the robots starting at $v_{x_i}^2$, for all $i = \{1, 2, \dots, n\}$ finish their motion at $v_{x_i}^4$. Since at least one station is necessary inside every variable gadget for the robots to move from $v_{x_i}^2$, for all i , the solution MIN-STATION problem on G_ϕ is n .

Now we assume that the MIN-STATION problem on G_ϕ has output n . We note again that it is necessary to place at least one station in every variable gadget to make a robot reach the target vertices in the variable gadgets.

Thus at most one station can be placed in every variable gadget. On the other hand, at least one station must be adjacent to the vertices present in each clause gadget to finish the motion of the robots present in the clause gadgets. Thus the motion can only be completed when at least one vertex corresponding to the literals of each clause has a station on them, implying the literal is assigned true for the clause. Hence, the lemma follows. $\square$

Combining Lemma 1 together with the maximum degree of G_ϕ , we get the following theorem.

Theorem 1. *MIN-STATION problem is NP-hard for graphs with maximum degree at most 6.*

Proof. The degree of each vertex present in the clause gadgets is 3. The degree of the terminal vertices in the variable gadgets is 2. The non-terminal vertices within the variable gadgets are adjacent to exactly two vertices from the variable gadget and, at most, four vertices from two clause gadgets. This follows from the fact that, in an instance of (3,3)-SAT, each literal occurs at most two times. If a variable has only positive or negative occurrences, it can be disregarded in the instance by assigning it a true value. Thus the maximum degree of a vertex in G_ϕ is at most 6. This fact along with Lemma 1 completes the proof. $\square$

4. Algorithms for Paths and Cycles

We present Algorithm 1 and 2 to solve the MIN-STATION problem on paths and cycles, respectively.

4.1. Algorithm for MIN-STATION on a path

We begin with an observation on the robots' motion on a path $P = \{v_1, v_2, \dots, v_n\}$. We assume the vertices are considered from left to right on the path for calculative brevity. Further, we assume that the initial and target positions of the robots are $\{s_1, s_2, \dots, s_m\}$ and $\{t_1, t_2, \dots, t_m\}$, respectively, from left to right. We recall that the movement capacity of each robot is denoted by r . We state the following lemma, which specifies the minimum number of stations required for a group of robots traveling in the same direction.

Lemma 2. *Let $1, 2, \dots, k$ be a group of k robots starting from $s_1, s_2, \dots, s_k \in P$, sorted from left to right. All of them need to reach a vertex $x \in P \setminus [s_1, s_k]$ and let x be lying on the right of s_k without loss of generality. Considering there is no preexisting station in the interval $[s_1, x]$, $\lfloor \frac{|x-s_1|}{r} \rfloor$ stations are necessary and sufficient for all the robots to reach x .*

Proof. We consider dividing the interval $[s_1, x]$ into $\lfloor \frac{|x-s_1|}{r} \rfloor$ portions starting from s_1 , to prove the necessary part of the claim. We note that the robot starting at s_1 must reach a station in every division of the interval $[s_1, x]$. This addresses the minimum number of stations required for the motion.

To prove the sufficiency of these stations to complete the motion, we consider the placement of the stations at the vertices $s_1 + r, s_1 + 2r, \dots, s_1 + \lfloor \frac{|x-s_1|}{r} \rfloor r$. We note that each robot has the capacity of r and as a result each of them can reach the immediate station on its right and hence the next station towards the right. Thus the motion is completed. $\square$

Lemma 2 helps to determine the minimum number of stations required when the paths of a group of robots intersect while they travel in a same direction. Here the *path of a robot* refers to the interval it travels on P from its starting position to its target position. Now we state a result when the paths of two robots intersect while they are moving in opposite directions of each other. We recall a motion refers to the complete movement of all the robots from their starting positions to their target positions.

Lemma 3. *Let the paths of two robots intersect while they are moving in opposite directions of each other. Then, the number of stations required for this motion is greater than or equal to the number of stations required in another motion where their paths do not intersect.*

Proof. Let the two robots starting at s_1 and s_2 be finishing at t_1 and t_2 , respectively, with s_1 being on the left of s_2 . Two intervals $[s_1, t_1]$ and $[s_2, t_2]$ can intersect each other in two ways: Either they intersect partially or one completely contains the other. We deal with both cases in the proof.

- **Case 1:** Let the intervals intersect partially. Let t_1 lie on the right of s_2 and t_2 lie on the left of s_1 , as depicted in Figure 2(a). Since each interval of length r starting from t_2 to t_1 must contain a station for the robots to reach the destinations, the minimum number of stations required to finish the motion of these two robots is at least $\lfloor \frac{|t_2-t_1|}{r} \rfloor$. However,

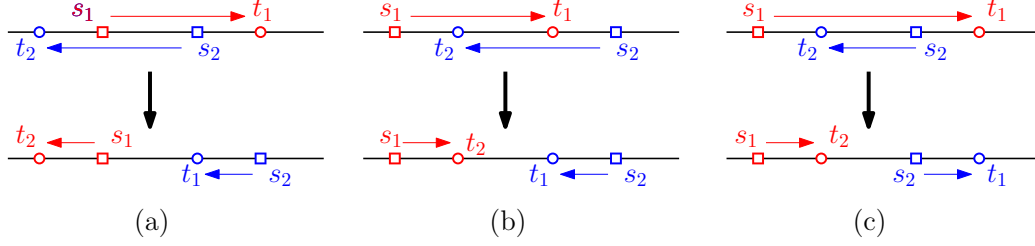


Figure 2: Target swapping, when two paths intersect and the robots are traveling in opposite directions of each other. The hollow squares and circles denote the starting and the target positions of the robots respectively. The movements of robot 1 and 2 are depicted in red and blue respectively in the colored figure.

when we consider the motion where the robots starting at s_1 and s_2 finishes at t_2 and t_1 respectively, the number of station required is $\lfloor \frac{|s_1-t_2|}{r} \rfloor + \lfloor \frac{|t_1-s_2|}{r} \rfloor \leq \lfloor \frac{|s_1-t_2|}{r} \rfloor + \lfloor \frac{|t_1-s_2|}{r} \rfloor + \lfloor \frac{|s_2-s_1|}{r} \rfloor \leq \lfloor \frac{|t_2-t_1|}{r} \rfloor$. A similar argument holds when t_1 lies on the left of s_2 , depicted in Figure 2(b).

- **Case 2:** Without loss of generality, let $[s_2, t_2]$ be completely contained inside $[s_1, t_1]$, as depicted in Figure 2(c). Then the minimum number of stations required for the robot at s_1 to reach t_1 is $\lfloor \frac{|t_1-s_1|}{r} \rfloor$. However, when we swap the target position of the robots, the number of stations required is $\lfloor \frac{|t_2-s_1|}{r} \rfloor + \lfloor \frac{|t_1-s_2|}{r} \rfloor \leq \lfloor \frac{|t_1-s_1|}{r} \rfloor$.

Combining these two cases the lemma holds. $\square$

Lemma 3 shows that in an optimal solution MIN-STATION problem on a path P , no two robots cross each other while moving in an opposite direction. Now we prove the following lemma that states the property of the motion that completes with the minimum number of charging stations.

Lemma 4. *The number of stations in MIN-STATION problem on a path P is minimized when the robot starting at s_i finishes its motion at t_i , for all $i \in \{1, 2, \dots, m\}$.*

Proof. We prove this lemma by induction on i . We begin with $i = 1$ as the induction base. Let the robot starting at s_1 do not finish its motion in t_1 but in some other target position t_o . Then some other robot finishes at t_1 to complete the motion of the system. Let s_o denote the starting position of this robot. We show that the number of stations required for this motion is higher than the number of stations required when the robots start at s_1 and

s_o and finish at t_1 and t_o , respectively. Now depending on the positions of the four vertices s_1, t_1, s_o and t_o on the path P , the following six cases arise.

- **Case 1:** The positions of the four vertices on P are t_1, t_o, s_1, s_o from the left to right on P . Then the minimum number of stations required to finish their motion is $\lfloor \frac{|s_o - t_1|}{r} \rfloor$, by virtue of Lemma 2. Now, this number of stations is also sufficient for the robots starting at s_o and s_1 to reach their targets at t_o and t_1 respectively. Thus the minimum number of stations required in the later motion is less than or equal to our assumed motion.
- **Case 2:** The positions of the four vertices on P are t_1, s_1, t_o, s_o from the left to right on P . If the robots starting at s_1 and s_o are finishing their motion at t_o and t_1 , respectively, then their target positions can be swapped by the Case 2 of Lemma 3.
- **Case 3:** The positions of the four vertices on P are t_1, s_1, s_o, t_o from the left to right on P . This case is addressed by the Case 1 of Lemma 3 as the paths intersect each other while the robots travel in an opposite direction.
- The remaining cases when the positions of the four vertices on P are (s_1, t_1, s_o, t_o) , (s_1, s_o, t_1, t_o) , and (s_1, t_1, t_o, s_o) from left to right. These cases are similar to Case 1, Case 2, and Case 3 of this lemma, stated above.

Now we consider the robot starting at s_{i+1} on P . The robots starting at $s_1, s_2, \dots, s_i$ are finishing their motion at $t_1, t_2, \dots, t_i$, respectively, by the induction hypothesis. The induction step follows from the following facts.

1. All the robots are of the same capacity. If there is any station placed on the right of s_{i+1} and a robot starting on the left of s_{i+1} , then the robot starting at s_{i+1} can reach the stations on its right.
2. If the robot starting at s_{i+1} reaches a target position t_o , then t_o lies on the right of t_{i+1} . If we denote by s_o , the starting position of the robot finishing at t_{i+1} , the motion of these two robots can be treated in a way similar to the induction base.

Thus the result follows. □

Input:

1. A path P of n vertices $\{v_1, v_2, \dots, v_n\}$ ordered from left to right.
2. A set of m starting and target positions $S = \{s_1, s_2, \dots, s_m\}$ and $T = \{t_1, t_2, \dots, t_m\}$, respectively, ordered from left to right.
3. A positive integer r .

Output: A minimum cardinality set $C \subset P$ of charging stations. $C = \phi;$

VCOUNTER=0, TCOUNTER=0;

for every $i = 1, 2, \dots, n$ **do** **if** $v_i \in S$ **then**

TCOUNTER++;

end **if** $v_i \in T$ **then**

TCOUNTER--;

end **if** TCOUNTER!=0 **then**

VCOUNTER++;

if VCOUNTER== r **then** $C = C \cup v_i;$

VCOUNTER=0;

end **end****end****return** $C;$ **Algorithm 1:** MIN-STATION-ON-PATH()

Now we devise an algorithm for the MIN-STATION problem on a path P with n vertices. Given the terminal positions of the robots from left to right on the path, we compute the minimum number of stations required for each robot using Lemma 4. However, calculating the value for each robot may require $O(n)$ time. Therefore, if we compute the values separately, the overall running time could increase to $O(n^2)$. Instead, we keep two counters from the starting and finishing positions of the robots while scanning the vertices of the path P from left to right. Thus we compute the number of stations required along with their positions on P with one single pass on the vertices of P . The steps are formally stated in Algorithm 1. We keep

a terminal counter (TCOUNTER) for the starting and target positions of the robots. We increase or decrease it by 1 when a starting or target position is scanned, respectively. The stations are placed in the interval spanned by two terminal vertices when the distance becomes equal to r . We keep track of this distance while scanning the vertices by another vertex counter (VCOUNTER).

Theorem 2. *MIN-STATION problem can be solved in $O(n)$ time for a path with n vertices and m robots, where $m \leq n$.*

Proof. Algorithm 1 returns a solution for the MIN-STATION problem and the correctness follows from Lemma 4. We note that the *for loop* in the algorithm processes each vertex on the path only once and the counters update the solution in constant time. Thus the running time of the algorithm is $O(n)$. $\square$

4.2. Algorithm for MIN-STATION for a cycle

We now design an algorithm for MIN-STATION problem for a cycle with n vertices. The property of an optimal motion of a MIN-STATION instance is as follows.

Lemma 5. *There exists at least one edge of the cycle in an optimal solution of a MIN-STATION instance, that is not traversed by any robot.*

Proof. We prove the lemma by showing that each motion addressing a MIN-STATION instance where each edge of the cycle is traversed by at least one robot, can be improved to another motion where some edge is not traversed and the number of stations required in the later one is lesser than the former one.

First, we consider the case when every edge is traversed by a robot and there is no path of the robots intersecting with each other. We note that this motion is redundant as every robot is starting from a target position. Hence, there is no need to move any of the robots to occupy different target positions, as depicted in Figure 3.

Now we consider the case where the paths of the robots are intersecting and every edge of the cycle is traversed by a robot. If there exist two paths intersecting each other where the robots are traveling in an opposite direction, then by Lemma 3 we can swap their target positions and there will be at least one edge with is not traversed by any robots.

So, we investigate the case when the paths intersect each other and the robots are traveling in a clockwise (say) direction. It is worth noting that in

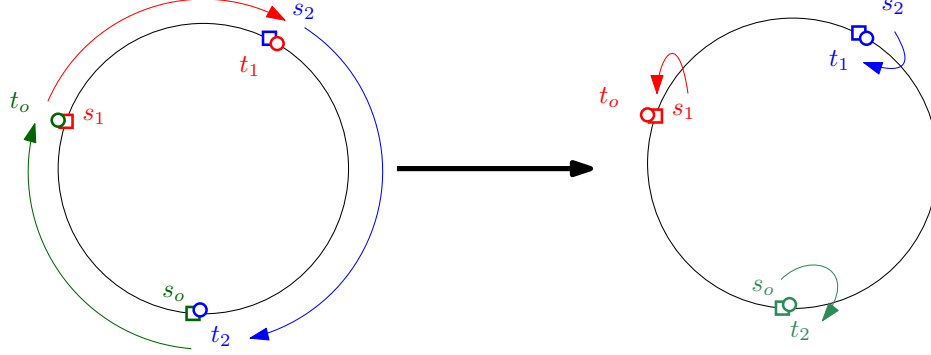


Figure 3: No movement is required in MIN-STATION for a cycle if all the edges are traversed by some robots and no path is intersecting with other paths.

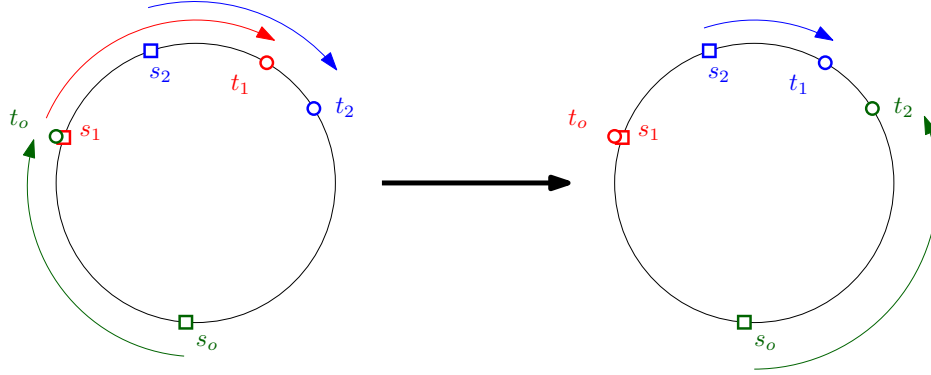


Figure 4: An edge in a cycle that is not traversed by any robot can be found, if there are intersecting paths of the robots present in the motion of a MIN-STATION instance.

our assumption, $\lfloor \frac{n}{r} \rfloor$ stations are required as every edge is being traversed by some robot. Let us consider the paths of two robots starting at s_i, s_{i+1} are intersecting with each other while they travel in the same direction towards their target positions t_i and t_{i+1} , respectively. Since all the edges of the cycle are traversed by some robot, s_i is also a target position t_o (say) for some other robot starting at s_o traveling clockwise. We improve the motion of these robots as follows, depicted in Figure 4.

- The robot starting at s_i remains at s_i occupying the target t_o .
- The robot starting at s_{i+1} moves to t_i . Hence the edges between s_i and s_{i+1} are not being traversed by any robots.

- The robot starting at s_o changes its direction and moves to t_{i+1} . This is possible with the existing stations in the interval from s_o to t_{i+1} in the anticlockwise direction as all of these edges are traversed by some robots.

Furthermore, we can argue in a similar way when more than two paths intersect. We consider two of them at a time and continue unless we get an edge of the cycle that is not traversed by a robot. This completes the proof by showing there is always one such path in an optimal solution of the MIN-STATION problem for a cycle. $\square$

Input:

1. A cycle G of n vertices $\{v_0, v_2, \dots, v_{n-1}\}$ sorted in clockwise order.
2. A set of m starting and target positions $S = \{s_1, s_2, \dots, s_m\}$ and $T = \{t_1, t_2, \dots, t_m\}$, respectively.
3. A positive integer r .

Output: A minimum cardinality set $C \subset G$ of charging stations.

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 $C = \{v_0, v_2, \dots, v_{n-1}\};$ 
for every  $i = 0, 1, 2, \dots, n - 1$  do
     $P = G \setminus \{\text{the edge between } v_i \text{ and } v_{(i+1)\%n}\};$ 
     $X = \text{MIN-STATION-ON-PATH}(P);$ 
    if cardinality of  $X \leq$  cardinality of  $C$  then
         $C = X;$ 
    end
end
return  $C;$ 

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Algorithm 2: MIN-STATION-ON-CYCLE()

Using Lemma 5 we devise Algorithm 2 for MIN-STATION problem for a cycle. The algorithm guesses an edge that is not traversed by any robots in an optimal solution and computes the MIN-STATION on the path constructed by removing the edge from the cycle. Then it returns the solution set C of the charging stations with the minimum cardinality among all the candidates. Since there are n edges in the cycle, Algorithm 2 returns an optimal solution in $O(n^2)$ time. Thus we have the following theorem.

Theorem 3. *MIN-STATION problem can be solved in $O(n^2)$ time for a cycle with n vertices.*

5. Conclusion

We conclude the article by pointing out some open problems arising from the work. We presented two algorithms for solving the MIN-STATION problem on paths and cycles and the problem remains NP-hard for general graphs. The immediate question that arises from the work is the complexity of the MIN-STATION problem for *path alike* graph classes such as trees or grids.

Question 1. *What is the complexity of the MIN-STATION problem for trees, grids, and planar graphs?*

Another avenue in this direction is considering the practical graph parameters [19] like pathwidth, treewidth, etc. It is relevant to study the fixed-parameter tractability of the problem under different parameters.

Question 2. *Is it possible to devise fixed-parameter tractable algorithms for MIN-STATION problem for graphs parameterized by the number of terminals, number of leaves, pathwidth, treewidth, etc.?*

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